

Xinyu Wang

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SUMMARY

Computer Science Ph.D. candidate with **7+ years** of machine learning research, including **3+ years** building operational ML systems. Develops deep learning and LLM methods spanning **representation learning**, **preference optimization**, and sequence modeling, with strengths relevant to retrieval, ranking, and personalization. Combines algorithm design with Python/PyTorch development and multi-GPU inference, delivering **billion-sample generation pipelines** and automated forecasting systems for utility partners.

EDUCATION

University of Connecticut

Sep 2019 – Expected Oct 2026

Ph.D. Candidate, Computer Science and Engineering

GPA: 3.97/4.00

University of Connecticut

Sep 2015 – May 2019

B.S.E., Computer Science & Engineering (Honors); B.A., Mathematics

GPA: 3.81/4.00

TECHNICAL SKILLS

Core: Python, PyTorch, C++, SQL/PostgreSQL

Modeling: Transformers, contrastive learning, preference optimization, LoRA, LLaMA

Systems: distributed inference, Linux/HPC, MPI, Slurm

EXPERIENCE

Laboratory of Machine Learning and Health Informatics, UConn

May 2017 – Present

Graduate Research Assistant

Storrs, CT

Semantic Representation Learning & Preference Optimization

- Developed SIGMA, a token-level training objective that aligns different strings representing the same molecule. Reduced generation-distribution error (FCD) by **9.9% and 19.4%** versus the strongest baselines across two QM9 molecular encodings, with no added inference cost.
- Extended SIGMA to property prediction, improving mean performance on all 6 molecular benchmarks while reducing sensitivity to equivalent input strings by **up to 44.9%**.
- Built and trained a low-rank output adapter on **1.89M** token-preference pairs to steer molecular generation with a frozen backbone. Combined gated updates to low-probability tokens with pairwise preference learning and reference-model regularization.

Foundation-Model Merging & Adaptation

- Developed DIAL, a training-free method that improves merged models by diagnosing and reweighting unreliable layers. Raised ViT-B/16 average accuracy from **77.6% to 82.3%** over the TIES merging baseline, without task examples or merge-time validation search.
- Designed piecewise layer scaling instead of imposing one linear rule across the network; outperformed optimized linear schedules in **14 of 15** model-merger settings. Extended DIAL from vision models to GPT-2 and Flan-T5 across 7 NLP tasks.

Scalable Generation

- Built an asynchronous multi-GPU pipeline that separates generation, token-level scoring, and evaluation, processing **1B+** molecular candidate samples in **6 days** on **8 V100 GPUs**.

Agent Learning

- Built replay-based action scorers that learn from states a language agent actually reaches, addressing gaps in offline training data. Increased pass rate from **56.1% to 78.1%** on 114 telecom tasks.

Structured Generation

- Developed a decoding method that checks molecular structure during generation and corrects token choices with reinforcement learning; reduced distribution error by 4.8% while preserving 99.5% valid outputs and 100% uniqueness over 10K samples.

UConn Outage Prediction Model, Eversource Energy Center

May 2023 – Present

Machine Learning Systems Engineer (Graduate Technician)

Storrs, CT

- Re-architected a legacy outage forecasting stack into an end-to-end ML pipeline serving utility partners across 4 regions and 4 daily forecast cycles. Integrated weather and utility data, feature preparation, regional predictions, and automated result delivery.
- Accelerated the WRF weather-modeling development pipeline through task parallelism and MPI tuning, cutting combined processing time by approximately **59%**, from about **4.5 hours to under 2 hours**.
- Resolved CPU-placement bottlenecks in model initialization with core binding and parallelized post-processing across 16 workers; matched-input tests confirmed consistent results across all 254 output files.
- Built failure recovery for 24/7 severe-weather forecasts: input-readiness gates, duplicate-job checks, and retry/resume logic with monitoring and alerts for data delays and system failures.
- Translated requirements from utility stakeholders, researchers, and engineers into an event-driven scheduler supporting 126-hour forecast horizons and up to 9 model initializations per severe-weather event.

PAPERS

- T. Zhu, F. Dou, **X. Wang**, J. Lu, J. Bi. Polyhedron Attention Module: Learning Adaptive-Order Interactions. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- **X. Wang**, F. Dou, J. Bi, M. Song. SIGMA: Semantic Identifier Grouping for Molecular Autoregression. Submitted to AAAI 2027.
- **X. Wang**, K. Deng, F. Dou, J. Bi, J. Lu. Beyond Linear Depth Scaling: Diagnosing Piecewise Layer Reliability for Data-Free Model Merging (DIAL). Submitted to NeurIPS 2026.
- **X. Wang**, J. Bi, M. Song. Leveraging Partial SMILES Validation Scheme for Enhanced Drug Design in Reinforcement Learning Frameworks. arXiv preprint.
- Z. Gao, **X. Wang**, B. Blumenfeld Gaines, X. Shi, J. Bi, M. Song. Fragment-based Deep Molecular Generation Using Hierarchical Chemical Graph Representation and Multi-resolution Graph Variational Autoencoder. *Molecular Informatics* 42(5), 2200215, 2023.
- S. Sahoo, C. Shende, M. Z. Hossain, P. Patel, Y. Niu, **X. Wang**, S. Ware, J. Bi. Cross-platform Prediction of Depression Treatment Outcome Using Location Sensory Data on Smartphones. *IEEE/ACM CHASE* 2025.
- S. Sahoo, C. Shende, M. Z. Hossain, P. Patel, **X. Wang**, M. I. Mahmud, J. Bi. Using Mobile Daily Mood and Anxiety Self-ratings to Predict Depression Symptom Improvement. *IEEE/ACM CHASE* 2024.
- C. Shende, S. Sahoo, S. Sam, P. Patel, R. Morillo, **X. Wang**, S. Ware, J. Bi. Predicting Symptom Improvement During Depression Treatment Using Sleep Sensory Data. *PACM IMWUT*, 2023.
- J. Lu, J. Sun, **X. Wang**, H. R. Kranzler, J. Gelernter, J. Bi. Inferring Phenotypes from Substance Use via Collaborative Matrix Completion. *BMC Systems Biology* 12, 15-27, 2018.
- J. Lu, J. Sun, **X. Wang**, H. R. Kranzler, J. Gelernter, J. Bi. Collaborative Phenotype Inference from Comorbid Substance Use Disorders and Genotypes. *IEEE BIBM* 2017.